

AER1416: Numerical Methods for Uncertainty Quantification — Assignment 3

Zouhair Adam Hamaimou Student #1004891986

Q1 — Non-Intrusive Polynomial Chaos Projection for P3**Problem Setup and Notation**

Problem P3 is the random linear system arising from the spatial discretisation of a 2-D stochastic diffusion equation,

$$\mathbf{A}(\boldsymbol{\theta}) \mathbf{u}(\boldsymbol{\theta}) = \mathbf{b}, \quad \mathbf{A}(\boldsymbol{\theta}) = \mathbf{A}_0 + \sum_{i=1}^d \theta_i \mathbf{A}_i, \quad \theta_i \sim \mathcal{U}[-3, +3] \text{ i.i.d.}, \quad (1)$$

with $\mathbf{A}_i \in \mathbb{R}^{n \times n}$, $\mathbf{b} \in \mathbb{R}^n$, $d = 3$ random inputs, and $n = 143$ interior degrees of freedom. We seek the componentwise mean $\boldsymbol{\mu}_u = \mathbb{E}[\mathbf{u}(\boldsymbol{\theta})]$ and variance $\boldsymbol{\sigma}_u^2 = \text{Var}[\mathbf{u}(\boldsymbol{\theta})]$ via a non-intrusive PC projection with a total-order Legendre basis of degree $p \in \{1, 2\}$.

Following the lecture notation, we write the truncated PC expansion as $\sum_{i=0}^P y_i \varphi_i(\boldsymbol{\theta})$, where $\{\varphi_i\}$ are the orthogonal basis functions and $\{y_i\}$ are the PC coefficients. Expectations under the input density are denoted $\langle \cdot \rangle$. Because $\mathbf{u} \in \mathbb{R}^n$ is vector-valued in space, the scalar coefficient y_i of the lecture becomes a vector $\mathbf{y}_i \in \mathbb{R}^n$ with components $y_{j,i}$ for each spatial DOF j . To avoid clashing with the problem's $n = 143$, we use $d = 3$ for the number of random inputs (what the lecture slides call n).

Mapping to Standard Legendre

Legendre polynomials are orthogonal on $[-1, 1]$ with density $\rho(\xi) = 1/2$, so we introduce the canonical variable

$$\xi_i = \theta_i/3, \quad \xi_i \sim \mathcal{U}[-1, +1], \quad (2)$$

and work with the transformed system

$$\left[\mathbf{A}_0 + 3 \sum_{i=1}^d \xi_i \mathbf{A}_i \right] \mathbf{u}(\boldsymbol{\xi}) = \mathbf{b}, \quad \rho(\boldsymbol{\xi}) = 2^{-d}.$$

Moments of $\mathbf{u}$ are invariant under this affine map, so every PC quantity below is computed in $\boldsymbol{\xi}$.

Generalised Polynomial Chaos Expansion

The lecture (slide 3) writes the truncated PC expansion as

$$u_j(\boldsymbol{\xi}) \approx \sum_{i=0}^P y_{j,i} \varphi_i(\boldsymbol{\xi}), \quad P+1 = \frac{(d+p)!}{d! p!}, \quad (3)$$

which for $d = 3$ gives $P+1 = 4$ at $p = 1$ and $P+1 = 10$ at $p = 2$. Each basis function is a tensor product of univariate Legendre polynomials,

$$\varphi_i(\boldsymbol{\xi}) = \prod_{j=1}^d L_{\alpha_j^{(i)}}(\xi_j),$$

indexed by a multi-index $\boldsymbol{\alpha}^{(i)} = (\alpha_1^{(i)}, \dots, \alpha_d^{(i)})$ satisfying $|\boldsymbol{\alpha}^{(i)}|_1 \leq p$. The basis is orthogonal,

$$\langle \varphi_i \varphi_{i'} \rangle = \delta_{ii'} \langle \varphi_i^2 \rangle, \quad \langle \varphi_i^2 \rangle = \prod_{j=1}^d \frac{1}{2\alpha_j^{(i)} + 1},$$

with the full $p = 2$ basis listed in Table 1.

Table 1: Total-order Legendre basis for Q1 ($d = 3$). The final column is $\langle \varphi_i^2 \rangle = \prod_{j=1}^d (2\alpha_j^{(i)} + 1)^{-1}$.

i	$\varphi_i(\xi_1, \xi_2, \xi_3)$	$\langle \varphi_i^2 \rangle$
<i>Order $p = 1$ (4 terms)</i>		
0	1	1
1	ξ_1	1/3
2	ξ_2	1/3
3	ξ_3	1/3
<i>Additional terms for $p = 2$ (10 terms total)</i>		
4	$\frac{1}{2}(3\xi_1^2 - 1)$	1/5
5	$\frac{1}{2}(3\xi_2^2 - 1)$	1/5
6	$\frac{1}{2}(3\xi_3^2 - 1)$	1/5
7	$\xi_1\xi_2$	1/9
8	$\xi_1\xi_3$	1/9
9	$\xi_2\xi_3$	1/9

Projection via Tensor-Product Gauss–Legendre Quadrature

By orthogonality (lecture, slide 8), the PC coefficients in (3) are

$$y_{j,i} = \frac{\langle u_j \varphi_i \rangle}{\langle \varphi_i^2 \rangle} = \frac{1}{\langle \varphi_i^2 \rangle} \int_{[-1,1]^d} u_j(\boldsymbol{\xi}) \varphi_i(\boldsymbol{\xi}) \rho(\boldsymbol{\xi}) d\boldsymbol{\xi}. \quad (4)$$

Because $\mathbf{u}(\boldsymbol{\xi})$ is a rational function of $\boldsymbol{\xi}$ (not a polynomial), we evaluate (4) with a tensor-product Gauss–Legendre rule of $q = p + 1$ nodes per dimension, which is exact for any polynomial integrand of total degree $\leq 2p + 1$. The total number of forward solves is

$$N_q = (p + 1)^d = \begin{cases} 8, & p = 1, \\ 27, & p = 2. \end{cases}$$

Denote the 1-D Gauss–Legendre nodes and weights by $\{\eta_r, \omega_r\}_{r=1}^{p+1}$, with weights normalised so that $\sum_r \omega_r = 1$ (absorbing the 1/2 density factor). The multivariate nodes $\boldsymbol{\xi}^{(\mathbf{r})} = (\eta_{r_1}, \dots, \eta_{r_d})$ and weights $W^{(\mathbf{r})} = \prod_k \omega_{r_k}$ give the discrete projection

$$y_{j,i} \approx \frac{1}{\langle \varphi_i^2 \rangle} \sum_{\mathbf{r}} W^{(\mathbf{r})} u_j(\boldsymbol{\xi}^{(\mathbf{r})}) \varphi_i(\boldsymbol{\xi}^{(\mathbf{r})}). \quad (5)$$

At each node, $u_j(\boldsymbol{\xi}^{(\mathbf{r})})$ is obtained by one $n \times n$ linear solve with the assembled matrix $\mathbf{A}_0 + 3 \sum_i \xi_i^{(\mathbf{r})} \mathbf{A}_i$.

Recovery of the Moments

Because $\varphi_0 \equiv 1$ and $\langle \varphi_i \rangle = 0$ for $i \geq 1$, the first two moments follow directly from the coefficients (lecture, slide 10):

$$\boxed{\mu_{u_j} = y_{j,0}, \quad \sigma_{u_j}^2 = \sum_{i=1}^P y_{j,i}^2 \langle \varphi_i^2 \rangle,} \quad (6)$$

which are evaluated componentwise across all $n = 143$ spatial DOFs.

Implementation

The accompanying MATLAB script `Q1.m` performs the following steps:

1. Load `P3.mat` (containing $\mathbf{A}_0, \dots, \mathbf{A}_3, \mathbf{b}$) and `P3grid.mat`.
2. Enumerate the total-order multi-index set $\{\boldsymbol{\alpha}^{(i)}\}_{i=0}^P$ for the required p .

3. Generate the $(p + 1)$ -point 1-D Gauss–Legendre nodes and weights via the Golub–Welsch construction of the Jacobi matrix, returning probabilistic weights (sum to 1).
4. Precompute the evaluation matrix $\Phi_{\mathbf{r},i} = \varphi_i(\boldsymbol{\xi}^{(\mathbf{r})})$ via Bonnet’s three-term recurrence, and the normalisation constants $\langle \varphi_i^2 \rangle$.
5. Loop over the N_q quadrature nodes: assemble the system matrix, solve, and accumulate the quadrature sums in (5).
6. Compute μ_u and σ_u^2 via (6) and visualise with the provided `PlotP3_3D` / `PlotP3_contour` routines. (A thin variant `PlotP3_contour_var` is used for the variance fields, since the stock contour routine hardcodes a level step of 0.005 that produces no contours at variance magnitudes of $\mathcal{O}(10^{-5})$.)

Results

Figures 1–4 show the mean and variance fields on the finite-element mesh at both orders, and Table 2 compares the statistics of the first interior DOF u_1 against the converged reference values from Assignments 1–2.

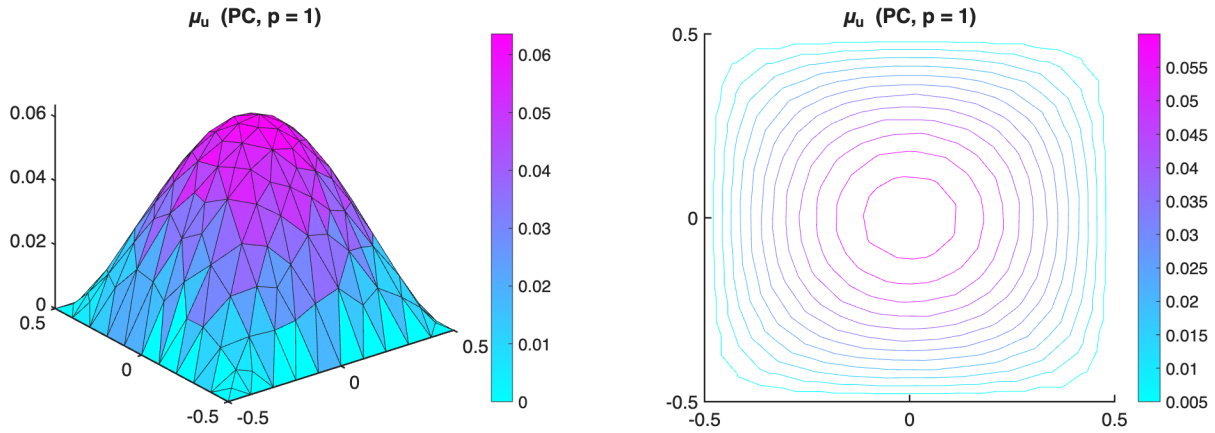


Figure 1: P3 mean field (non-intrusive PC, $p = 1$, 8 solves): 3-D surface (left) and contour (right).

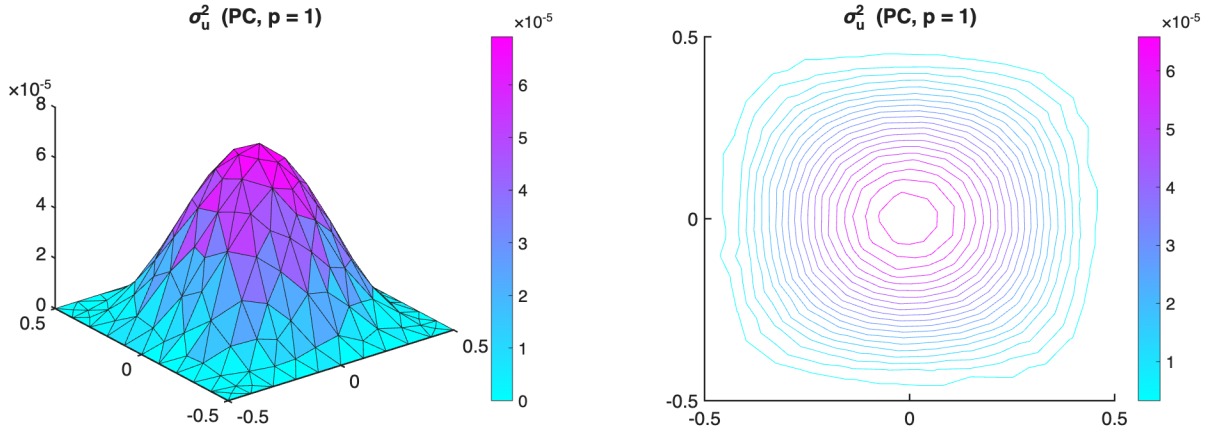


Figure 2: P3 variance field (non-intrusive PC, $p = 1$, 8 solves): 3-D surface (left) and contour (right).

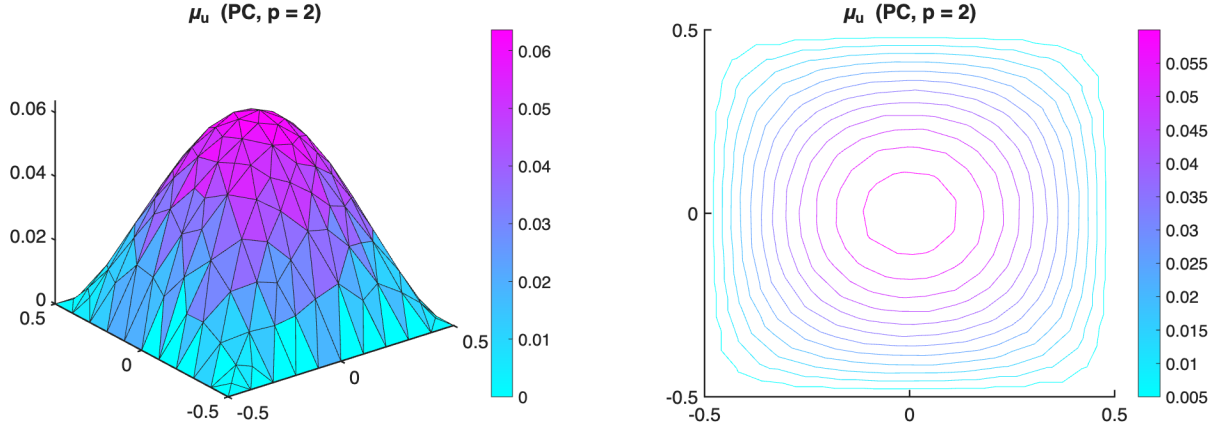


Figure 3: P3 mean field (non-intrusive PC, $p = 2$, 27 solves): 3-D surface (left) and contour (right).

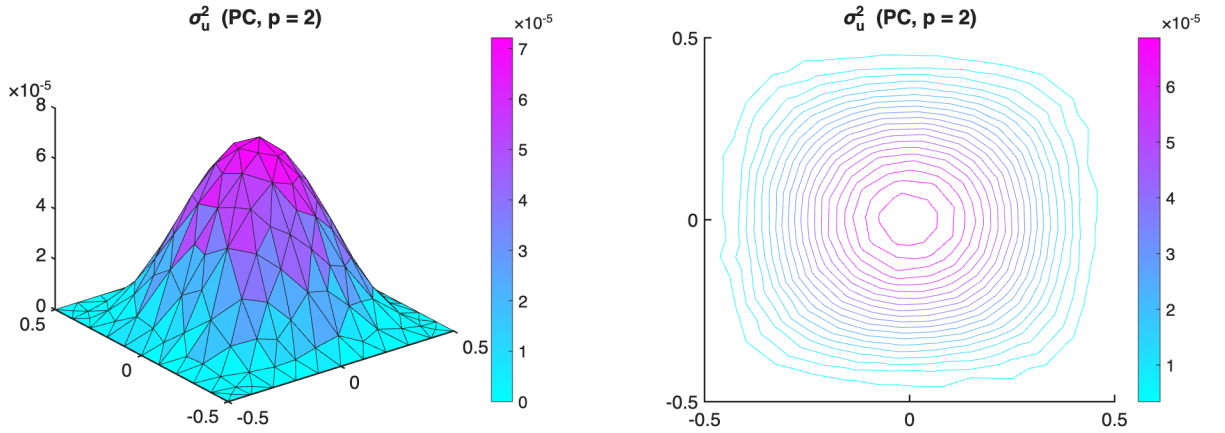


Figure 4: P3 variance field (non-intrusive PC, $p = 2$, 27 solves): 3-D surface (left) and contour (right).

Table 2: Comparison of u_1 statistics across all methods studied in the course. PC values are from this assignment; sparse-CC, Sobol, LHS and MC values are taken from the converged runs reported in Assignments 1–2. “Cost” is the number of forward linear-system solves — for the sampling methods, the sample size at which the method has stabilised to three significant figures on this problem.

Method	Cost	μ_{u_1}	$\sigma_{u_1}^2$
PC, $p = 1$	8	6.3590×10^{-2}	6.914×10^{-5}
PC, $p = 2$	27	6.3607×10^{-2}	7.214×10^{-5}
Sparse CC (level 3)	69	6.361×10^{-2}	7.221×10^{-5}
Sobol QMC ($N = 5,000$)	5,000	6.361×10^{-2}	7.224×10^{-5}
LHS ($N = 5,000$)	5,000	6.361×10^{-2}	7.220×10^{-5}
Monte Carlo ($N = 50,000$)	50,000	6.366×10^{-2}	7.222×10^{-5}

Accuracy. Taking the sparse-CC level-3 estimate ($\mu = 6.361 \times 10^{-2}$, $\sigma^2 = 7.221 \times 10^{-5}$) as the reference, the PC mean at $p = 1$ is already within 0.03% and the variance within 4.3%. Raising the order to $p = 2$ drops the mean error to 0.005% and the variance error to 0.10%, matching the reference to four significant figures. The 4% variance gap at $p = 1$ is expected: a first-order expansion captures only the linear sensitivities

$\partial u_j / \partial \xi_i$, whereas with $\theta \in [-3, 3]$ the quadratic contributions $y_{j,i}^2 \langle \varphi_i^2 \rangle$ coming from the $|\alpha^{(i)}|_1 = 2$ terms are non-negligible. Adding the $p = 2$ basis captures the dominant quadratic coefficients and accounts for almost all of the remaining gap.

Cost. Table 2 also highlights the efficiency gap between PC and the sampling or sparse-quadrature benchmarks on this smooth, low-dimensional problem. The $p = 2$ expansion reaches converged accuracy with only 27 solves, roughly $2.5\times$ cheaper than sparse CC at level 3 (69 points) and more than three orders of magnitude cheaper than the Sobol, LHS, and MC runs used in Assignments 1–2. The mean and variance fields in Figures 1–4 are visually indistinguishable from those produced by LHS and Sobol at $N = 2 \times 10^5$ in Assignment 2, exhibiting the same centred bell-shaped mean with both fields vanishing at the Dirichlet boundary. For problems with smooth parameter dependence and modest input dimension, non-intrusive PC is a strong candidate in this efficiency–accuracy regime.

Q2 — Intrusive Stochastic Galerkin Projection for P2

Problem Statement and Notation

Problem P2 is the forced, damped single-degree-of-freedom oscillator,

$$m(\theta) \ddot{u}(t, \theta) + c(\theta) \dot{u}(t, \theta) + k(\theta) u(t, \theta) = F_0(\theta) \cos(\omega(\theta) t), \quad u(0) = \dot{u}(0) = 0, \quad (7)$$

with $\theta = (m, c, k, F_0, \omega)$, each component uniform and independent: $m \sim \mathcal{U}[0.9, 1.1]$, $c \sim \mathcal{U}[0.8, 1.2]$, $k \sim \mathcal{U}[10, 12]$, $F_0 \sim \mathcal{U}[0.9, 1.1]$, $\omega \sim \mathcal{U}[0.1, 0.2]$. We map each parameter to the canonical interval $[-1, 1]$ via $\alpha(\theta) = \alpha_0 + \alpha_1 \xi_\alpha$, with $\alpha_0 = \frac{1}{2}(\alpha^{\max} + \alpha^{\min})$, $\alpha_1 = \frac{1}{2}(\alpha^{\max} - \alpha^{\min})$, so that

$$\xi_m, \xi_c, \xi_k, \xi_F, \xi_\omega \sim \mathcal{U}[-1, 1] \text{ i.i.d.},$$

and write $\xi = (\xi_m, \xi_c, \xi_k, \xi_F, \xi_\omega) \in [-1, 1]^5$ with joint density $\rho(\xi) = 2^{-5}$.

First-Order Legendre PC Basis

A total-order Legendre basis of degree $p = 1$ in $d = 5$ inputs contains $P + 1 = \frac{(d+p)!}{d! p!} = 6$ terms. We order them as

$$\varphi_0 = 1, \quad \varphi_1 = \xi_m, \quad \varphi_2 = \xi_c, \quad \varphi_3 = \xi_k, \quad \varphi_4 = \xi_F, \quad \varphi_5 = \xi_\omega.$$

The orthogonality relation is $\langle \varphi_r, \varphi_s \rangle = h_r \delta_{rs}$ with $h_0 = \langle \varphi_0^2 \rangle = 1$ and $h_r = \langle \varphi_r^2 \rangle = \frac{1}{3}$ for $r \geq 1$.

Each random parameter is expanded in this basis (the expansion is exact since $p = 1$ captures a scalar uniform input):

$$m = m_0 + m_1 \varphi_1, \quad c = c_0 + c_1 \varphi_2, \quad k = k_0 + k_1 \varphi_3, \quad F_0 = F_{00} + F_{01} \varphi_4, \quad \omega = \omega_0 + \omega_1 \varphi_5, \quad (8)$$

where, for example, $m_0 = 1.0$, $m_1 = 0.1$; $c_0 = 1.0$, $c_1 = 0.2$; $k_0 = 11$, $k_1 = 1$; $F_{00} = 1.0$, $F_{01} = 0.1$; $\omega_0 = 0.15$, $\omega_1 = 0.05$.

Stochastic Galerkin Projection

Assume the solution admits a truncated PC expansion

$$u(t, \theta) \approx \sum_{i=0}^P u_i(t) \varphi_i(\xi), \quad (9)$$

where the modes $\{u_i(t)\}_{i=0}^P$ are deterministic functions of time. Substituting (9) into (7) yields a residual $R(t, \xi)$; the Galerkin condition requires R to be orthogonal to every basis function,

$$\langle R(t, \xi), \varphi_n \rangle = 0, \quad n = 0, 1, \dots, P. \quad (10)$$

Expanding each coefficient in (7) via (8) and using (9),

$$R(t, \boldsymbol{\xi}) = \sum_i \ddot{u}_i(t) m(\boldsymbol{\xi}) \varphi_i + \sum_i \dot{u}_i(t) c(\boldsymbol{\xi}) \varphi_i + \sum_i u_i(t) k(\boldsymbol{\xi}) \varphi_i - F_0(\boldsymbol{\xi}) \cos(\omega(\boldsymbol{\xi}) t).$$

Taking the inner product with φ_n and invoking linearity produces

$$\sum_{i=0}^P M_{ni} \ddot{u}_i(t) + \sum_{i=0}^P C_{ni} \dot{u}_i(t) + \sum_{i=0}^P K_{ni} u_i(t) = \mathcal{F}_n(t), \quad n = 0, \dots, P, \quad (11)$$

with

$$M_{ni} = \langle m(\boldsymbol{\xi}) \varphi_i \varphi_n \rangle = m_0 \langle \varphi_i \varphi_n \rangle + m_1 \langle \varphi_1 \varphi_i \varphi_n \rangle, \quad (12)$$

$$C_{ni} = \langle c(\boldsymbol{\xi}) \varphi_i \varphi_n \rangle = c_0 \langle \varphi_i \varphi_n \rangle + c_1 \langle \varphi_2 \varphi_i \varphi_n \rangle, \quad (13)$$

$$K_{ni} = \langle k(\boldsymbol{\xi}) \varphi_i \varphi_n \rangle = k_0 \langle \varphi_i \varphi_n \rangle + k_1 \langle \varphi_3 \varphi_i \varphi_n \rangle, \quad (14)$$

$$\mathcal{F}_n(t) = \langle F_0(\boldsymbol{\xi}) \cos(\omega(\boldsymbol{\xi}) t) \varphi_n \rangle. \quad (15)$$

Computing the Triple Products

The only non-zero pieces are $\langle \varphi_i \varphi_n \rangle = h_n \delta_{ni}$ and the triple products $e_{rni} = \langle \varphi_r \varphi_n \varphi_i \rangle$ with $r \in \{1, \dots, 5\}$. Because the basis is tensor-separable and each non-trivial φ_r depends on a distinct coordinate, the expectation factorises across coordinates, with each factor being a 1-D integral of a product of Legendre polynomials against the uniform density on $[-1, 1]$. For the first-order basis used here, every non-constant φ_r ($r = 1, \dots, 5$) is a single coordinate ξ_r , so these 1-D integrals can be evaluated directly without appealing to the general Legendre triple-product formula. Given one factor $\varphi_r = \xi_r$ inside $e_{rni} = \langle \varphi_r \varphi_n \varphi_i \rangle$, the remaining two factors φ_n and φ_i must combine to produce a ξ_r -dependent piece (otherwise the ξ_r -integral vanishes by odd symmetry), and also must leave no unintegrated factor in any other coordinate. Both conditions are met only when $\{\varphi_n, \varphi_i\} = \{1, \varphi_r\}$, giving

$$\langle \varphi_r \varphi_0 \varphi_r \rangle = \langle \varphi_r \varphi_r \varphi_0 \rangle = \mathbb{E}[\xi_r^2] = \frac{1}{3},$$

while all other first-order triple products vanish by odd symmetry in ξ_r or by mismatched coordinates. Consequently the Galerkin matrices are mostly diagonal with a single off-diagonal row/column pair coupling each mode to the zeroth mode:

$$\mathbf{M} = \begin{bmatrix} m_0 & m_1/3 & 0 & 0 & 0 & 0 \\ m_1/3 & m_0/3 & 0 & 0 & 0 & 0 \\ 0 & 0 & m_0/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & m_0/3 & 0 & 0 \\ 0 & 0 & 0 & 0 & m_0/3 & 0 \\ 0 & 0 & 0 & 0 & 0 & m_0/3 \end{bmatrix}, \quad (16)$$

with an analogous structure for $\mathbf{C}$ (coupling at indices 0–2) and $\mathbf{K}$ (coupling at indices 0–3). Both $\mathbf{M}$ and $\mathbf{K}$ are symmetric positive definite, since the corresponding random parameters are strictly positive and the diagonal entries are all positive.

Forcing Term Projection

The forcing $f(t, \boldsymbol{\xi}) = F_0(\boldsymbol{\xi}) \cos(\omega(\boldsymbol{\xi}) t)$ is not polynomial in $\boldsymbol{\xi}$ because ω enters through a transcendental function, but the Galerkin projections $\mathcal{F}_n(t) = \langle F_0(\boldsymbol{\xi}) \cos(\omega(\boldsymbol{\xi}) t) \varphi_n \rangle$ are still one-dimensional integrals in ξ_ω and ξ_F that admit a closed form. Using the angle-addition identity $\cos((\omega_0 + \omega_1 \xi_\omega) t) = \cos(\omega_0 t) \cos(\omega_1 t \xi_\omega) - \sin(\omega_0 t) \sin(\omega_1 t \xi_\omega)$ and evaluating the two ξ_ω -integrals against $\rho(\xi_\omega) = 1/2$ on $[-1, 1]$,

$$\frac{1}{2} \int_{-1}^1 \cos(\omega_1 t \xi) d\xi = \text{sinc}(\omega_1 t), \quad \frac{1}{2} \int_{-1}^1 \xi \sin(\omega_1 t \xi) d\xi = \frac{\text{sinc}(\omega_1 t) - \cos(\omega_1 t)}{\omega_1 t},$$

with $\text{sinc}(x) = \sin(x)/x$, we obtain the exact projections

$$\begin{aligned} \mathcal{F}_0(t) &= F_{00} \cos(\omega_0 t) \text{sinc}(\omega_1 t), \\ \mathcal{F}_1(t) &= \mathcal{F}_2(t) = \mathcal{F}_3(t) = 0, \\ \mathcal{F}_4(t) &= \frac{1}{3} F_{01} \cos(\omega_0 t) \text{sinc}(\omega_1 t), \\ \mathcal{F}_5(t) &= -F_{00} \sin(\omega_0 t) \frac{\sin(\omega_1 t) - \omega_1 t \cos(\omega_1 t)}{(\omega_1 t)^2}. \end{aligned} \quad (17)$$

Only modes 0, 4, 5 are directly forced; modes 1, 2, 3 are excited through the off-diagonal couplings in $\mathbf{M}$, $\mathbf{C}$, $\mathbf{K}$.

Remark on the small- $\omega_1 t$ limit. A first-order Taylor expansion $\cos((\omega_0 + \omega_1 \xi_\omega)t) \approx \cos(\omega_0 t) - \omega_1 t \xi_\omega \sin(\omega_0 t)$ recovers (17) in the limit $\omega_1 t \rightarrow 0$, since $\text{sinc}(\omega_1 t) \rightarrow 1$ and, using $\sin x - x \cos x \sim x^3/3$ as $x \rightarrow 0$, $\mathcal{F}_5(t) \rightarrow -\frac{1}{3} F_{00} \omega_1 t \sin(\omega_0 t)$. On the present problem, $\omega_1 = 0.05$ and $T = 10$ give $\omega_1 T = 0.5$, so the sinc factors deviate from unity by roughly 4%, a non-negligible correction that the full projection (17) captures exactly.

Summary of the Galerkin System

Collecting $\mathbf{u}(t) = (u_0(t), \dots, u_5(t))^\top \in \mathbb{R}^6$ and $\mathcal{F}(t) = (\mathcal{F}_0(t), \dots, \mathcal{F}_5(t))^\top$, the stochastic Galerkin projection yields a coupled deterministic second-order linear ODE system,

$$\mathbf{M} \ddot{\mathbf{u}}(t) + \mathbf{C} \dot{\mathbf{u}}(t) + \mathbf{K} \mathbf{u}(t) = \mathcal{F}(t), \quad \mathbf{u}(0) = \dot{\mathbf{u}}(0) = \mathbf{0}. \quad (18)$$

The moments of the scalar response follow directly from the coefficients:

$$\mathbb{E}[u(t, \boldsymbol{\theta})] = u_0(t), \quad \text{Var}[u(t, \boldsymbol{\theta})] = \sum_{i=1}^5 h_i u_i(t)^2 = \frac{1}{3} \sum_{i=1}^5 u_i(t)^2. \quad (19)$$

Efficient Numerical Solution Scheme

We propose the following scheme to solve (18). We emphasise we are *not* implementing it here; the points below describe how one would do so efficiently.

Step 1 — Reduction to first-order form. Define the state $\mathbf{y}(t) = (\mathbf{u}(t); \dot{\mathbf{u}}(t))^\top \in \mathbb{R}^{12}$. Equation (18) becomes

$$\dot{\mathbf{y}}(t) = \mathbf{A} \mathbf{y}(t) + \mathbf{g}(t), \quad \mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1} \mathbf{K} & -\mathbf{M}^{-1} \mathbf{C} \end{bmatrix}, \quad \mathbf{g}(t) = \begin{bmatrix} \mathbf{0} \\ \mathbf{M}^{-1} \mathcal{F}(t) \end{bmatrix}. \quad (20)$$

Step 2 — Exploit the sparsity of $\mathbf{M}$, $\mathbf{C}$, $\mathbf{K}$. Each of these 6×6 matrices has at most 8 non-zero entries and is symmetric positive definite. In particular, $\mathbf{M}$ decomposes into a 2×2 block (for indices $\{0, 1\}$) and a 4×4 diagonal block (for indices $\{2, 3, 4, 5\}$). We precompute a Cholesky factor $\mathbf{M} = \mathbf{L}\mathbf{L}^\top$ once; every inversion of $\mathbf{M}$ at run time costs $\mathcal{O}(6)$ via a single triangular solve.

Step 3 — Timestepping. The natural frequency of the deterministic part is $\omega_n = \sqrt{k_0/m_0} \approx 3.3$ rad/s and all Galerkin modes share the same time scale; the system is non-stiff over the required horizon. An explicit 4-stage Runge-Kutta (RK4) scheme is therefore an appropriate default choice. A step size Δt of order $(\omega_n)^{-1}/10 \approx 0.03$ s is a reasonable starting heuristic that resolves the dominant oscillation with a safety margin; a more careful choice would run an a-posteriori error estimator (e.g. Richardson extrapolation over Δt or an embedded RK4(5) pair) and adapt accordingly. Each RK4 step costs four triangular solves with the precomputed Cholesky factor plus sparse matrix-vector products, i.e. $\mathcal{O}(1)$ flops per step for a 6×6 system. Over the horizon $[0, 10]$ s, the total cost is negligible compared with running the same time-marching scheme over thousands of Monte Carlo samples.

Step 4 — Moment post-processing. At every output time, evaluate the mean and variance from the modal coefficients using (19). No additional quadrature or sampling is required because the PC expansion analytically encodes all moment information.

Step 5 — Extensions. Two avenues extend the scheme beyond $p = 1$:

- Increasing the polynomial order to $p = 2$ yields $P + 1 = 21$ modes, which captures quadratic variation of the response in the parameters (e.g. ξ_k^2, ξ_ω^2) that $p = 1$ omits. The forcing projection (17) generalises by using the same identities against each higher-degree Legendre polynomial in ξ_ω , yielding closed-form sinc-type coefficients at every order.
- For larger input dimensions or higher orders, the coupling pattern of $\mathbf{M}, \mathbf{C}, \mathbf{K}$ becomes denser but remains highly structured. Block-sparse Cholesky or Krylov-based ODE integrators would help control the per-step cost as the Galerkin system grows.

Q3 — Probabilistic Numerics and Uncertainty in Computations

The paper by Hennig, Osborne, and Girolami [3] argues that classical deterministic numerical methods can be reinterpreted as instances of Bayesian statistical inference: assign a prior to the unknown mathematical object, a likelihood to the data produced by running the algorithm, and take the posterior mean as the point estimate. Under well-chosen priors the classical result is reproduced exactly, and the posterior covariance provides a built-in error model. For uncertainty quantification (UQ), this closes a gap that the standard toolbox of this course does not: every method we have studied produces a point estimate and treats the remaining numerical error as either negligible or quantified by an unrelated mechanism. The sections below develop the mathematical content of the reinterpretation (Q3a), its consequences for ODE integration (Q3b), and its impact on the polynomial chaos expansions used in Q1 and Q2 (Q3c).

(a) Classical Numerical Methods as Bayesian Inference

Every deterministic numerical algorithm maps a finite set of observations about a latent object z (an integral, a linear-system solution, an ODE trajectory) to a point estimate $\hat{z}$. The PN reinterpretation makes this an inference problem by specifying a prior $p(z)$, a likelihood $p(\mathcal{D} | z)$ relating evaluations $\mathcal{D} = \{(x_i, y_i)\}$ to z , and taking the posterior mean $\hat{z} = \mathbb{E}[z | \mathcal{D}]$ as the estimate. The posterior covariance $\text{Cov}[z | \mathcal{D}]$ is a byproduct of the same computation, requires no extra evaluations, and serves as the epistemic counterpart of the aleatoric uncertainty that UQ methods propagate.

Quadrature. For $F = \int_a^b f(x) dx$, a GP prior $f \sim \mathcal{GP}(0, k)$ induces a Gaussian prior on F by linearity. Conditioning on $\mathbf{y} = (f(x_1), \dots, f(x_N))^T$ yields

$$p(F | \mathbf{y}) = \mathcal{N}\left(\int_a^b \mathbf{k}(x, \mathbf{X}) \mathbf{K}^{-1} \mathbf{y} dx, \int_a^b \int_a^b [k(x, x') - \mathbf{k}(x, \mathbf{X}) \mathbf{K}^{-1} \mathbf{k}(\mathbf{X}, x')] dx dx'\right), \quad (21)$$

with $\mathbf{K}_{ij} = k(x_i, x_j)$. The posterior mean is a quadrature rule with weights $\mathbf{w} = \int_a^b \mathbf{k}(x, \mathbf{X}) dx \mathbf{K}^{-1}$. With suitable kernel choices and node placements, classical rules re-emerge: for example, a piecewise-linear (Brownian-bridge) kernel with equispaced nodes reproduces the trapezoidal rule. More generally, Bayesian quadrature provides a unifying framework in which classical rules are special cases, though not every classical rule is recovered by an obvious kernel. The posterior variance in (21) is the kernel-based counterpart of a worst-case error bound: under fixed kernel hyperparameters it depends only on the node locations (through $\mathbf{K}$), not on the observed $\mathbf{y}$. Dependence on the data enters when the kernel hyperparameters are themselves inferred, e.g. by marginal-likelihood maximisation, which converts the posterior variance into a genuinely data-adaptive uncertainty estimate.

Linear algebra and ODEs. For an SPD system $\mathbf{A}\mathbf{x} = \mathbf{b}$, placing a matrix-variate Gaussian prior on $\mathbf{H} = \mathbf{A}^{-1}$ and conditioning on exact matrix-vector observations $\mathbf{y}_i = \mathbf{A}\mathbf{s}_i$ yields a posterior mean iterate

$\hat{\mathbf{x}}_M = \hat{\mathbf{H}}_M \mathbf{b}$ that coincides with the M -th conjugate-gradient step, with covariance tracking the part of $\mathbf{A}^{-1}$ not yet explored by the Krylov subspace. Similarly, for the IVP $\dot{\mathbf{y}} = \mathbf{f}(t, \mathbf{y})$, a Gauss-Markov prior (e.g. an integrated Wiener process) combined with a likelihood enforcing $\dot{\mathbf{y}}(t_{n,i}) = \mathbf{f}(t_{n,i}, \tilde{\mathbf{y}}_{n,i})$ at stage nodes reproduces classical explicit Runge-Kutta methods as the posterior mean, with per-step covariance available essentially for free [3].

Why this matters for UQ. UQ pipelines decompose the total error into an *aleatoric* part (from randomness in the inputs $\boldsymbol{\theta}$) and an *epistemic* part (from finite solver resolution). Writing $u(\boldsymbol{\theta})$ for the exact response and $\hat{u}(\boldsymbol{\theta})$ for the numerical approximation, the total variance of a QoI ϕ decomposes as

$$\text{Var}[\phi(\hat{u}(\boldsymbol{\theta}))] = \underbrace{\text{Var}_{\boldsymbol{\theta}}[\mathbb{E}[\phi(\hat{u}) \mid \boldsymbol{\theta}]]}_{\text{aleatoric (parameter)}} + \underbrace{\mathbb{E}_{\boldsymbol{\theta}}[\text{Var}_{\text{num}}(\phi(\hat{u}) \mid \boldsymbol{\theta})]}_{\text{epistemic (solver)}}. \quad (22)$$

Classical UQ assumes the second term is negligible and replaces $\hat{u}(\boldsymbol{\theta}^{(n)})$ with $u(\boldsymbol{\theta}^{(n)})$ in the MC sum. When the solver tolerance is comparable to the parameter-induced spread, this assumption breaks and the reported MC confidence interval under-states the true uncertainty. PN makes Var_{num} directly computable in a probability model compatible with the aleatoric one, allowing rational allocation of effort between sampling and solver accuracy.

(b) Application: Differential Equations

I focus on ODE integration, the forward model for Problem P2 of this course. A classical explicit Runge-Kutta (RK) method advances $t_n \rightarrow t_{n+1}$ by evaluating $\mathbf{f}$ at s stage points and forming a linear combination with Butcher weights (b_i), yielding a point estimate $\hat{\mathbf{y}}_{n+1}$ with local truncation error $\mathcal{O}(\Delta t^{q+1})$ for order q . The global error is a deterministic Gronwall bound with no distributional structure.

The PN reinterpretation. Hennig et al. [3] place a q -times integrated Wiener prior on $\mathbf{y}(t)$, whose sample paths lie in C^{q-1} and match the regularity that order- q RK tacitly assumes. Conditioning on the stage observations $\dot{\mathbf{y}}(t_{n,i}) = \mathbf{f}(t_{n,i}, \tilde{\mathbf{y}}_{n,i})$ for $i = 1, \dots, s$ yields a Gaussian posterior on $\mathbf{y}(t_{n+1})$ whose mean reproduces the classical RK update and whose covariance $\boldsymbol{\Sigma}_{n+1}$ scales as Δt^{2q+1} , matching the local-truncation-error variance up to a calibration constant. $\boldsymbol{\Sigma}_{n+1}$ is computed within the same Kalman recursion that advances $\hat{\mathbf{y}}_n$, so propagating it over $[0, T]$ yields a full trajectory posterior $\mathcal{N}(\hat{\mathbf{y}}(t), \boldsymbol{\Sigma}(t))$ at negligible extra cost.

Algorithmic consequences. Three design decisions change once a per-step posterior covariance is available. First, *step-size control*: classical adaptive RK forms a local-error estimate by embedding two updates of orders q and $q+1$ and taking their difference, which is a heuristic surrogate for the posterior standard deviation that the PN machinery computes directly; replacing the embedded norm by $\text{tr } \boldsymbol{\Sigma}_{n+1}$ gives a controller that targets a calibrated uncertainty rather than a mismatch between two arbitrarily related methods. Second, *stage-node selection*: classical stages maximise accuracy order, but one may instead choose $(t_{n,i})$ to minimise $\text{tr } \boldsymbol{\Sigma}_{n+1}$, which for linear problems has a closed form and for nonlinear ones reduces to a low-dimensional offline optimisation. Third, *coupling to UQ*: when the ODE is embedded in a Monte Carlo loop over $\boldsymbol{\theta}$ as in Assignments 1 and 2, propagating $\boldsymbol{\Sigma}(t; \boldsymbol{\theta})$ through the sampler recovers the full variance decomposition (22) at essentially no extra cost; the MC confidence interval at each t is replaced by a composite interval reflecting both sampling error and solver posterior uncertainty. The result is a solver that returns a distribution rather than a single curve, with the reported QoI uncertainty accounting for both error sources and the allocation between “more samples” and “tighter solver” decidable quantitatively rather than by heuristic.

(c) Synergy: Probabilistic Numerics and Polynomial Chaos Expansions

Of the methods developed in this course I focus on polynomial chaos expansions, which are the central technique of this assignment and the one whose epistemic error structure is most naturally expressed in Bayesian terms. Q1 reported point estimates of the P3 moments at $p = 1$ and $p = 2$ and justified stopping at $p = 2$ by visual comparison of the two, without ever quantifying the residual truncation error. Q2 went further

in the other direction: the intrusive Galerkin derivation produced a deterministic 6-dimensional coupled ODE system, but since implementation was not required, no credible interval or convergence diagnostic for the truncation was ever discussed. PN closes both gaps by revealing PC as a Bayesian linear model in an orthogonal basis, with truncation acting as a prior and collocation values (or Galerkin residuals) as data.

A PC estimate actually carries three distinct errors that classical PCE conflates: (i) *truncation* of modes with $|\boldsymbol{\alpha}^{(i)}|_1 > p$, controlled by the Audouze-Nair a-priori bound $\|u - \Pi_p u\|_{L^2_p} \leq C_d p^{-k} \|u\|_{H^k_p}$ with incomputable constants; (ii) *projection-integration* error from replacing $y_i = \langle u \varphi_i \rangle / \langle \varphi_i^2 \rangle$ by its quadrature approximation (exact up to degree $2p + 1$ for the tensor-product Gauss-Legendre rule used in Q1, at cost $\mathcal{O}((p + 1)^d)$); and (iii) the *aleatoric variance* $\sigma_u^2 = \sum_{i \geq 1} y_i^2 \langle \varphi_i^2 \rangle$ that is the statistical output, not an error. PN separates them.

PCE as a Bayesian linear model. The cleanest PN reinterpretation puts a Gaussian prior directly on the coefficient vector $\mathbf{y} = (y_0, \dots, y_p)^\top$:

$$p(\mathbf{y}) = \mathcal{N}(\mathbf{0}, \boldsymbol{\Lambda}_p), \quad \boldsymbol{\Lambda}_p = \text{diag}(\lambda_0^2, \dots, \lambda_p^2). \quad (23)$$

The prior variances λ_i^2 encode expected decay of the spectrum: for an integrand of regularity H^k , the Audouze-Nair estimate suggests $\lambda_i^2 \propto |\boldsymbol{\alpha}^{(i)}|_1^{-2k}$, which is exactly the “sparse in the chaos basis” structure discussed in the lecture slides (slide 14). Given m collocation evaluations $\mathbf{u}_{\text{obs}} = (u(\boldsymbol{\xi}^{(1)}), \dots, u(\boldsymbol{\xi}^{(m)}))^\top$ and the evaluation matrix $\Phi_{qi} = \varphi_i(\boldsymbol{\xi}^{(q)})$, the likelihood

$$p(\mathbf{u}_{\text{obs}} | \mathbf{y}) = \mathcal{N}(\Phi \mathbf{y}, \sigma_\varepsilon^2 \mathbf{I}) \quad (24)$$

accommodates both quadrature noise (set $\sigma_\varepsilon \rightarrow 0$ for noise-free collocation as in Q1) and genuine observation error when the forward model u is itself estimated from Monte Carlo or from a noisy simulation. The posterior is Gaussian,

$$p(\mathbf{y} | \mathbf{u}_{\text{obs}}) = \mathcal{N}(\boldsymbol{\mu}_y, \boldsymbol{\Sigma}_y), \quad \boldsymbol{\Sigma}_y = (\sigma_\varepsilon^{-2} \Phi^\top \Phi + \boldsymbol{\Lambda}_p^{-1})^{-1}, \quad \boldsymbol{\mu}_y = \sigma_\varepsilon^{-2} \boldsymbol{\Sigma}_y \Phi^\top \mathbf{u}_{\text{obs}}. \quad (25)$$

When $\sigma_\varepsilon \rightarrow 0$ and $\boldsymbol{\Lambda}_p \rightarrow \infty$ (uninformative prior, noise-free observations), $\boldsymbol{\mu}_y$ reduces to the ordinary least-squares (OLS) estimator $(\Phi^\top \Phi)^{-1} \Phi^\top \mathbf{u}_{\text{obs}}$. This OLS fit is *not* equivalent to the Gauss-Legendre projection formula (5) used in Q1: the projection $y_i = \langle u \varphi_i \rangle / \langle \varphi_i^2 \rangle$ discretised on quadrature nodes gives $\hat{y}_i = \sum_q W_q u_q \varphi_i(\boldsymbol{\xi}_q) / \langle \varphi_i^2 \rangle$, which is the *weighted* normal-equations solution with weight diagonal $\mathbf{W} = \text{diag}(W_q)$ and basis-norm scaling, i.e. the solution of $(\Phi^\top \mathbf{W} \Phi) \mathbf{y} = \Phi^\top \mathbf{W} \mathbf{u}_{\text{obs}}$. For the tensor-product rule on $m = (p + 1)^d$ nodes the basis is orthogonal under $\mathbf{W}$ (so $\Phi^\top \mathbf{W} \Phi$ is diagonal), and the solution coincides exactly with (5). To recover Q1’s projection inside the PN framework, one therefore replaces the isotropic likelihood (24) by a heteroscedastic version $p(\mathbf{u}_{\text{obs}} | \mathbf{y}) = \mathcal{N}(\Phi \mathbf{y}, \sigma_\varepsilon^2 \mathbf{W}^{-1})$. Under this weighted likelihood with flat prior, the MAP estimate is the Q1 Gauss-Legendre projection; under the isotropic likelihood, it is OLS, which is the regression (rather than projection) flavour of non-intrusive PC.

Posterior on the moments. Because $\mu_u = y_0$ and $\sigma_u^2 = \sum_{i \geq 1} y_i^2 \langle \varphi_i^2 \rangle$ are linear and quadratic functions of $\mathbf{y}$ respectively, the posterior (25) induces closed-form posteriors on both moments:

$$\mathbb{E}[\mu_u | \mathbf{u}_{\text{obs}}] = [\boldsymbol{\mu}_y]_0, \quad \text{Var}[\mu_u | \mathbf{u}_{\text{obs}}] = [\boldsymbol{\Sigma}_y]_{00}, \quad (26)$$

$$\mathbb{E}[\sigma_u^2 | \mathbf{u}_{\text{obs}}] = \sum_{i=1}^P ([\boldsymbol{\mu}_y]_i^2 + [\boldsymbol{\Sigma}_y]_{ii} \langle \varphi_i^2 \rangle). \quad (27)$$

The variance posterior is non-central χ^2 rather than Gaussian, but its first moment is available in closed form and already reveals something important: the *variance of the variance estimate* depends on the diagonal of the posterior covariance $\boldsymbol{\Sigma}_y$, not just on the mean coefficients. In Assignment 1 the MC variance estimator at $N = 200,000$ carried a 99% CI half-width of $\pm 5.9 \times 10^{-7}$ on $\sigma_{u_1}^2 \approx 7.2 \times 10^{-5}$. The PCE counterpart would be a credible interval on the same quantity derived from (27) at no additional model evaluations.

Principled truncation via model evidence. Classical PCE picks p by eyeballing when moments stabilise across orders (how I argued for $p = 2$ in Q1). PN replaces this with Bayesian model selection via the log marginal likelihood

$$\log p(\mathbf{u}_{\text{obs}} | p) = -\frac{1}{2} \mathbf{u}_{\text{obs}}^\top \mathbf{K}_p^{-1} \mathbf{u}_{\text{obs}} - \frac{1}{2} \log |\mathbf{K}_p| - \frac{m}{2} \log 2\pi, \quad \mathbf{K}_p = \Phi \Lambda_p \Phi^\top + \sigma_\varepsilon^2 \mathbf{I}. \quad (28)$$

Maximising (28) implements automatic Occam-razor balancing: the data-fit term rewards accuracy, the log-determinant penalises overly flexible models. This extends to the dimension-adaptive and sparse PC variants on slides 14–18 of the course lectures, where the evidence ranks candidate index sets and selects which $\alpha^{(i)}$ to retain.

Applying the framework to the Q2 Galerkin scheme. The framework applies naturally to the intrusive scheme derived in Q2. Although the forcing projection (17) is exact at $p = 1$, the $p = 1$ truncation of the solution expansion $u(t, \xi) \approx \sum_{i=0}^5 u_i(t) \varphi_i(\xi)$ still discards all higher-order coupling of the response to products such as $\xi_m \xi_\omega$, ξ_k^2 , and so on, and classical Galerkin has no mechanism to quantify the resulting truncation error other than rerunning at a higher order. A probabilistic Galerkin variant places a Gaussian prior on the coefficient trajectories $\mathbf{u}(t)$ and models the neglected higher-order modes as latent noise driving the observed modes, yielding a time-dependent posterior covariance $\Sigma_u(t)$ that (26)–(27) propagate into credible intervals for $\mu_u(t)$ and $\sigma_u^2(t)$. Interval widths grow as the coupling to truncated modes accumulates over time and contract as the basis order is increased, giving an explicit time-resolved error budget for a scheme whose deterministic form offers none.

Theoretical and practical gains. The PN reformulation offers several immediate gains. Theoretically, the incomputable Audouze-Nair a-priori bound is replaced by a computable posterior variance (conditional on the prior kernel, or fully data-adaptive if the hyperparameters are inferred); model selection over candidate truncations $\{p_k\}$ and candidate index sets $\{\mathcal{I}_k\}$ proceeds by ranking the log evidence (28) across alternatives rather than by ad-hoc convergence inspection, which is a principled discrete model comparison (not a single convex optimisation) but still replaces visual inspection with a scalar criterion; and the non-intrusive and intrusive variants of Q1 and Q2 unify as MAP estimates under the Bayesian linear model (23)–(24) with different likelihoods (collocation or weighted-collocation vs. Galerkin residual), so switching between them becomes a modelling choice rather than a change of algorithm. Practically, the diagonal of Σ_y in (25) gives credible intervals at cost $\mathcal{O}((P+1)^3)$, which for $P+1 = 10$ in Q1 is effectively free; the next collocation node can be chosen to minimise expected posterior variance on the target QoI, giving a principled dimension-adaptive rule for $d > 3$ where tensor-product collocation is infeasible; a sparsity-inducing prior (e.g. automatic relevance determination, treating λ_i^2 as hyperparameters inferred from the data) is exactly the Bayesian analogue of the compressive-sensing PC discussed on slide 14 of the lectures, with the posterior covariance additionally identifying which selected modes are well-resolved; and because truncation uncertainty (Σ_y) and parameter uncertainty (the density of θ) are both Gaussian measures on compatible spaces, the variance decomposition (22) closes in closed form.

Summary. PN offers a probabilistic generalisation of the PCE framework used in this assignment. The coefficient-projection formulas of Q1 (under a weighted likelihood) and the Galerkin update of Q2 re-emerge as MAP estimates under a natural Bayesian linear model; Σ_y provides a credible interval for coefficients and downstream moments whose dependence on data enters through hyperparameter inference; the marginal likelihood (28) gives a principled criterion for ranking candidate truncations; and the variance decomposition (22) makes numerical and parameter uncertainty commensurable. For someone who just spent an assignment picking p by visual inspection of a convergence table, the PN reformulation is the missing piece that would let me report not just “ $p = 2$ is sufficient” but “ $p = 2$ is sufficient to reduce the posterior standard deviation of $\sigma_{u_1}^2$ below a prescribed tolerance”.

References

- [1] D. Xiu and G. E. Karniadakis. The Wiener–Askey polynomial chaos for stochastic differential equations. *SIAM J. Sci. Comput.*, 24(2):619–644, 2002.

- [2] D. Lucor and G. E. Karniadakis. Adaptive generalized polynomial chaos for nonlinear random oscillators. *SIAM J. Sci. Comput.*, 26(2):720–735, 2004.
- [3] P. Hennig, M. A. Osborne, and M. Girolami. Probabilistic numerics and uncertainty in computations. *Proc. Royal Society A*, 471:20150142, 2015.
- [4] A. O’Hagan. Monte Carlo is fundamentally unsound. *J. Royal Statistical Society, Series D*, 36(2/3):247–249, 1987.